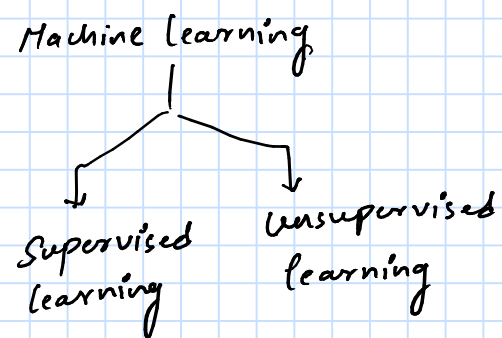
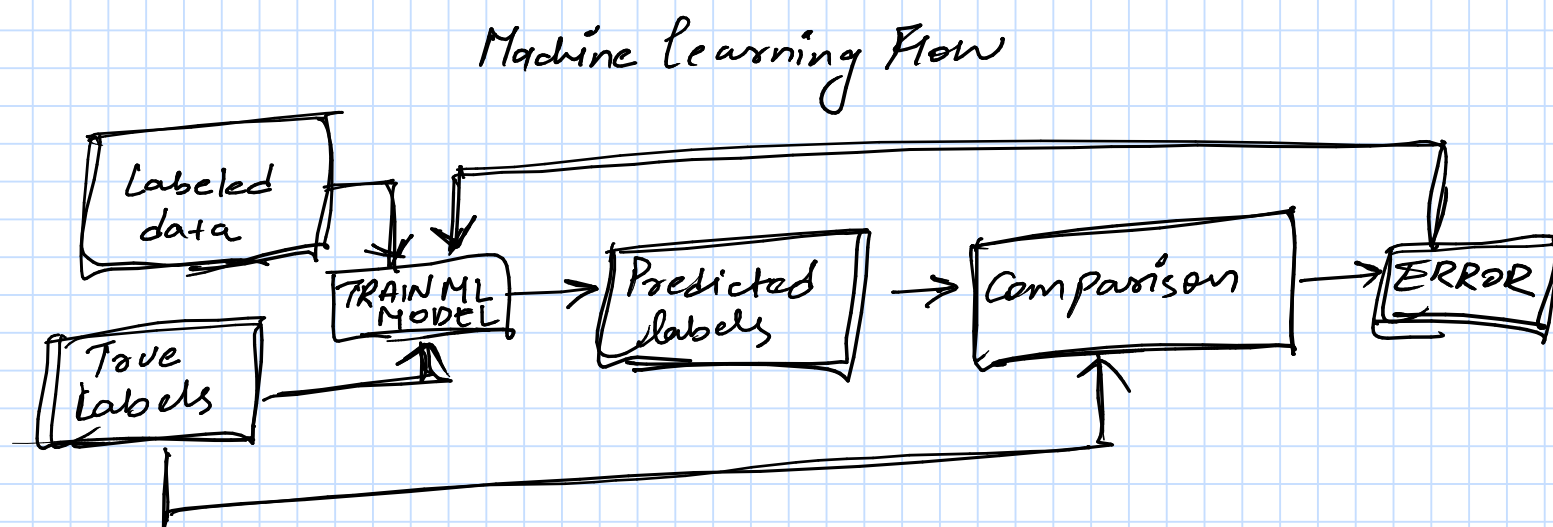




Supervised learning: ML model learns based on examples of data and labels.

E.g. Regression - labels will be in continuous numerical values
 Classification - labels will be in the form of classes
 0, 1, yes/no



Basic Work flow of Machine learning
Work flow.

- Step I Gather the data
- Step II Data quality assurance, normalization, feature engineering
- Step III Split the data into training and testing.
- Step IV Use training data to train the model

training data ~ 70%
 testing data ~ 30% } → Random Selection

Regression

Predict a continuous numerical value

E.g. House price Prediction
 Insurance Premium Prediction

Classification

Predict a category/class from predefined categories.

E.g.: Image object classification
 human/car/bike } MULTILABEL CLASSIFICATION
 Email spam detection
 spam/not a spam } BINARY CLASSIFICATION

UNSUPERVISED LEARNING

NO LABELS

ONLY HELPS IN IDENTIFYING THE PATTERNS

o UN-SUPERVISED LEARNING IS THE TYPE OF ML ALGORITHMS IN WHICH ML MODEL IDENTIFIES PATTERNS IN THE DATA

o It automatically identifies group of similar examples.

E.g: Clustering Models.

⇒ Group customers with similar location, salary etc.

⇒ Group similar customer complaints

Linear Regression

o Linear Regression is a supervised machine learning algorithm for regression problems.

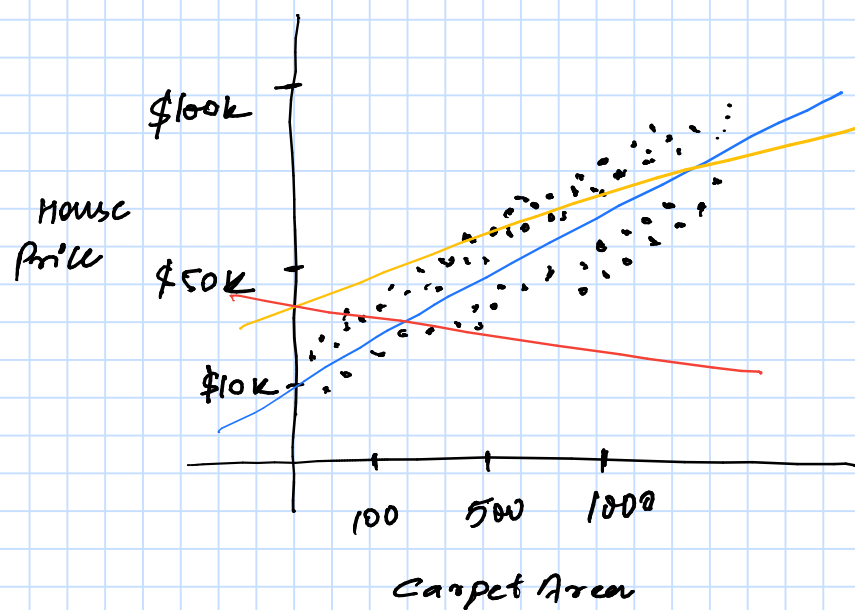
Regression ⇒ Continuous numerical variables.

Linear ⇒ Assuming that the label is linearly dependent on the feature.

E.g: House Price = $\theta_0 + \theta_1 \times \text{area}$ Equation of the line
Goal is to find the optimal value of θ_0 and θ_1 , which will result in minimum error
↓
Difference between Predicted and true values

Goal is to always capture general trend in the training data.

Individual data points can be noisy and hence the noise is ignored.



- $H \rightarrow$ House Price
- $A \rightarrow$ Carpet Area

Assuming H & A are linear

$$H = \theta_0 + \theta_1 \times A \sim y = mx + c$$

$\theta_0 \rightarrow$ Intercept (c)

$\theta_1 \rightarrow$ Slope (m)

Blue line represents the relationship between carpet area and house price.

What is Linear Regression

It is Supervised learning algorithm - Labelled data points are needed to train a model.

Linear - It can learn linear relationship between input variables (features) and output variable.

Regression - It predicts continuous numerical values as output.

Example: Predict House price based on carpet area in sq. feet.

- Linear regression with single feature and single output is called a simple linear regression.

$$Y_i = \theta_0 + \theta_1 X_i + \epsilon_i$$

For each observation $i = 0, 1, 2, 3, \dots, n$

- Y_i is vector of response variable
- X_i is vector of independent variable

$y \rightarrow$ dependent variable
Response variable
Predicted variable

• θ_0 is y intercept

• θ_1 is slope of line - coefficients

• ϵ_i is error term which represents random error created by any factors contributing to value of y which is not represented by x .

i	x_i	y_i	$\hat{y}_i$	ϵ_i
0	x_0	y_0	$\theta_0 + \theta_1 x_0$	$y_0 - (\theta_0 + \theta_1 x_0)$
1	x_1	y_1	$\theta_0 + \theta_1 x_1$	$y_1 - (\theta_0 + \theta_1 x_1)$
2	x_2	y_2	$\theta_0 + \theta_1 x_2$	$y_2 - (\theta_0 + \theta_1 x_2)$
3	x_3	y_3	$\theta_0 + \theta_1 x_3$	$y_3 - (\theta_0 + \theta_1 x_3)$

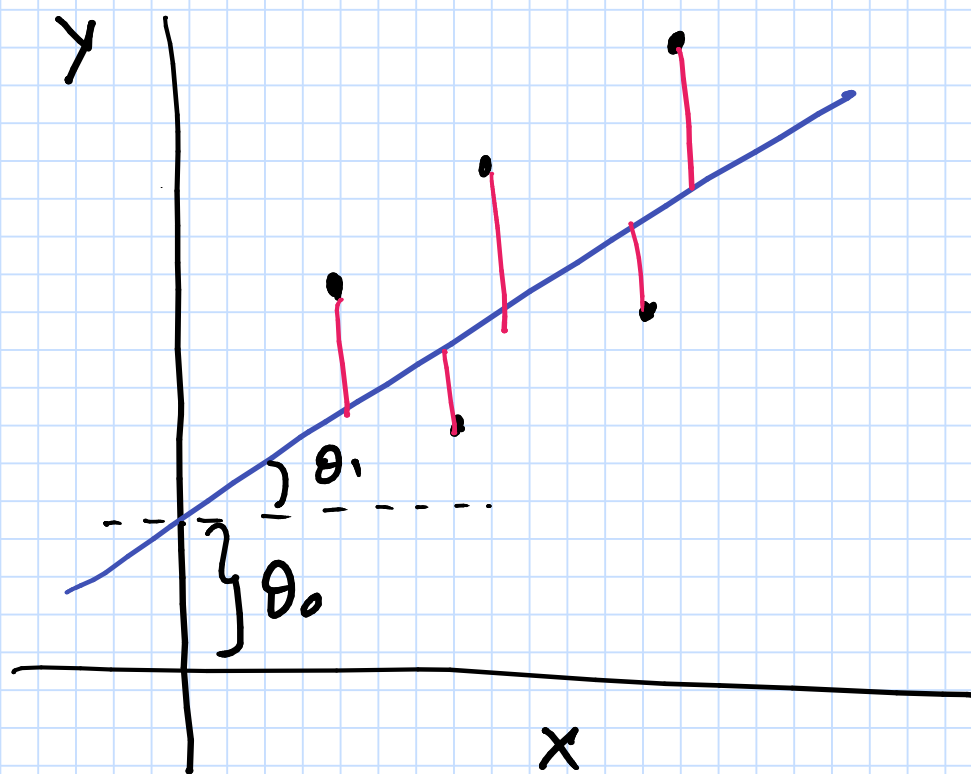
Residuals

- Difference between predicted value and actual value.
- Residuals represent errors in prediction.

$$\text{Residual} = \hat{y}_i - y_i$$

$y_i \rightarrow$ Actual

$\hat{y}_i \rightarrow$ Predicted by Model.



Assumptions of Linear Regression.

Linearity

- There should be linear relationship between independent variables and the dependent variables.

No Multicollinearity

- Independent variables should not be correlated with each other.
- When the independent variables are correlated, it is difficult to isolate the impact of a single factor on the dependent variable which can lead to unexpected results.

Independence of Residuals

- Residuals should not exhibit a systematic pattern or correlation

Ex: % Residuals increase with price.

- If there is a pattern in the residuals, it suggests that there may be some information left in the data that your model has not captured.

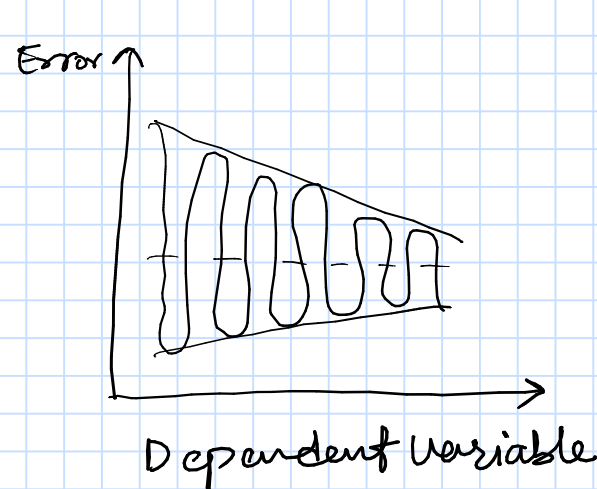
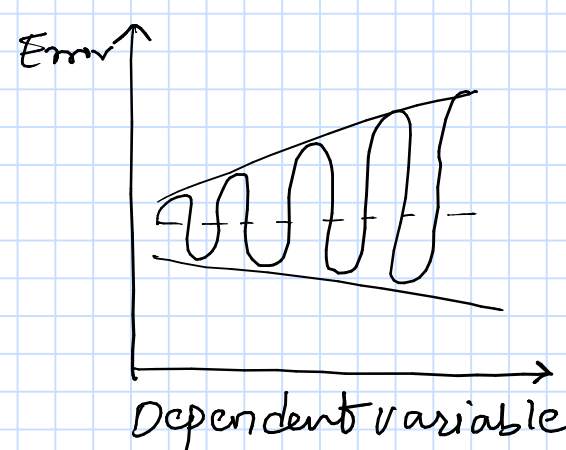
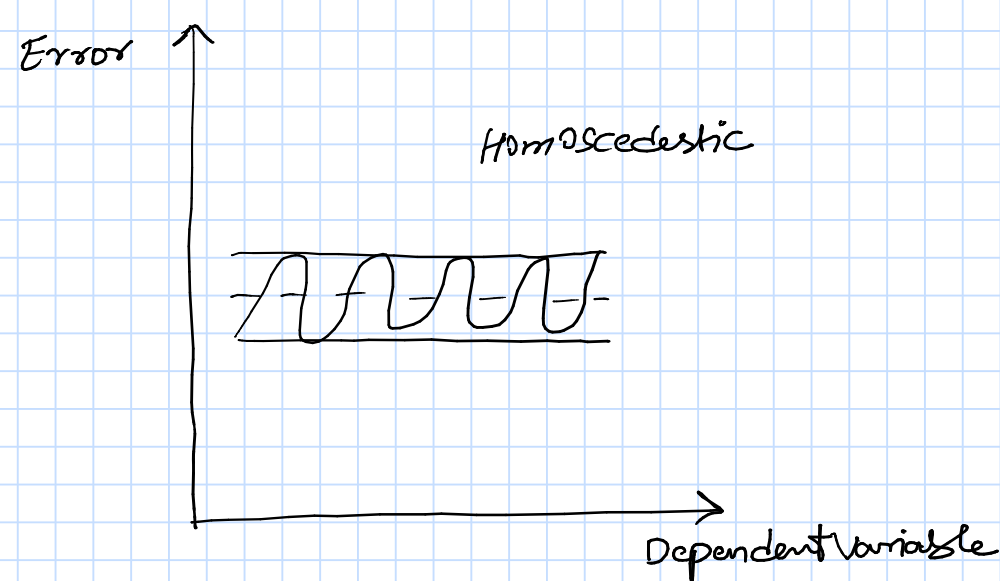
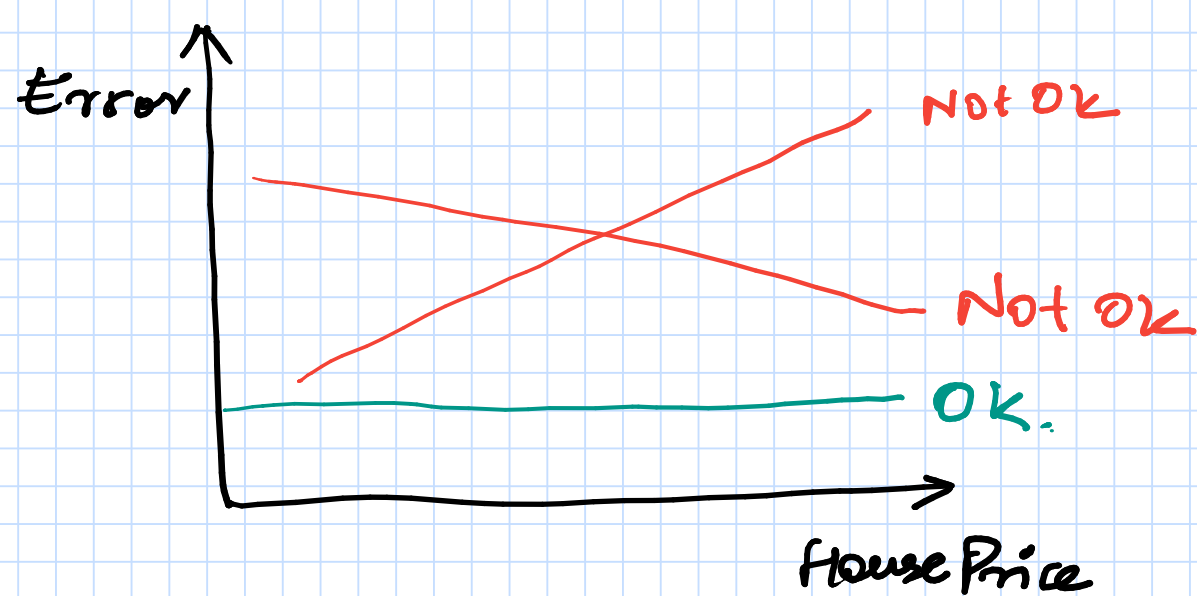
Error Distribution follows standard normal distribution
Mean is Zero.

Constant Variance (Homoscedasticity)

Homoscedasticity in a model means that the error has constant variance along the values of the dependent variable.

Normality

Residuals are normally distributed.



Heteroscedastic.

COST FUNCTION

$$y_i = \theta_0 + \theta_1 x_i$$

How to estimate values of $\theta_0 + \theta_1$?

We should choose those values of θ_0 and θ_1 which minimize error between predicted value of model and actual value.

Cost Function

$$J(\theta) = \frac{1}{2M} \sum_{i=0}^{M-1} \epsilon_i^2$$

$\epsilon \rightarrow$ Residual/Error

$M \rightarrow$ Number of samples

Number of Rows

$$\Rightarrow J(\theta) = \frac{1}{2M} \sum_{i=0}^{M-1} (\hat{y}_i - y_i)^2$$

$\hat{y}_i \rightarrow$ Predicted value.

$$\Rightarrow J(\theta) = \frac{1}{2M} \sum_{i=0}^{M-1} (\theta_0 + \theta_1 x_i - y_i)^2$$

MSE Mean Squared Error

Average Squared Error

We have to minimize the error function.
or Cost function.

Why square?

$\Rightarrow$ To Ensure all errors are positive
and to amplify the impact of larger errors.

Why divide by $2M$?

- Normalize the sum of squared errors
making it mean squared error.

- This ensures cost function does not depend
on the number of training examples/ rows
and allows comparison across different datasets.

COST FUNCTION & GRADIENT DESCENT

→ Find Optimum Parameters

Least Squares Regression

$$J(\theta) = \frac{1}{2M} \sum_{i=0}^{M-1} (\theta_0 + \theta_1 x_i - y_i)^2$$

If $\theta_0 = 0$ i.e., fitted line passes through $(0,0)$
then we will have to choose that value of θ_1
which gives minimum value of $J(\theta)$

Gradient Descent.

$$\frac{\partial J(\theta)}{\partial \theta} = \frac{1}{M} \sum_{i=0}^{M-1} (\theta_1 x_i - y_i) x_i \quad \text{J} \rightarrow \text{Gradient}$$

1) Calculate the gradient of the Cost function

$$\text{Cost function} = J(\theta) = \frac{1}{M} \sum_{i=0}^{M-1} (\theta_1 x_i - y_i) x_i$$

$$\Rightarrow \frac{\partial J(\theta)}{\partial \theta_1} = \frac{1}{M} \sum_{i=0}^{M-1} (\theta_1 x_i - y_i) x_i$$

$$\theta_{\text{new}} = \theta_{\text{old}} - \alpha \frac{\partial J}{\partial \theta} \bigg|_{\theta_{\text{old}}} \quad \text{For } \alpha = 0.1$$

$\alpha \rightarrow \text{learning rate}$

Learning Rate
 α is provided by us.

Iteration Stopping Criteria

(1) We have reached the predecided number of iterations.

(2) $|\theta_{\text{new}} - \theta_{\text{old}}| < \text{threshold}$
i.e. there are no

updates in θ values.

(3) $\frac{\partial J}{\partial \theta} < \text{threshold}$

Gradient Descent Flow

→ Start with some θ_0, θ_1

→ Keep changing θ_0 and θ_1 to reduce $J(\theta_0, \theta_1)$
until we hopefully end up at a minimum.

Gradient Descent Algorithm

$$\theta_i := \theta_i - \alpha \frac{\partial J(\theta)}{\partial \theta_i} \quad \text{Repeat until Convergence} \\ \text{For } i=0 \text{ and } i=1$$

- $\alpha \rightarrow$ learning rate
- Parameters are updated based on the ^{Derivative} gradient of cost function and learning rate.
- θ_0 and θ_1 should be updated simultaneously after each gradient calculation.

$$\theta_i \rightarrow \theta_0, \theta_1 \quad i \in \{0, 1\}$$

$$J(\theta_0, \theta_1) = \frac{1}{2N} \sum_{i=0}^{N-1} (\theta_0 + \theta_1 x_i - y_i)^2$$

$$\frac{\partial J}{\partial \theta_0} = \frac{1}{N} \sum_{i=0}^{N-1} (\theta_0 + \theta_1 x_i - y_i) = \frac{1}{N} \sum_{i=0}^{N-1} (\theta_0 + \theta_1 x_i - y_i)$$

$$\frac{\partial J}{\partial \theta_1} = \frac{1}{N} \sum_{i=0}^{N-1} (\theta_0 + \theta_1 x_i - y_i) x_i \\ = \frac{1}{N} \sum_{i=0}^{N-1} (\theta_0 + \theta_1 x_i - y_i) x_i$$

$$\boxed{\theta_{\text{new}} = \theta_{\text{old}} - \frac{\partial J(\theta)}{\partial \theta_{\text{old}}}} \rightarrow \text{Calculate for both } \theta_0 \text{ and } \theta_1$$

If $f(\theta) = \theta^2$

Evaluation

Then $\frac{\partial f(\theta)}{\partial \theta} = 2\theta$

i.e. $f(\theta) = \theta^n$

$$\frac{\partial f(\theta)}{\partial \theta} = n\theta^{n-1}$$

If $f(\theta) = (a\theta + b)^n$

Then $\frac{\partial f(\theta)}{\partial \theta} = n(a\theta + b)^{n-1} [a + 0]$

$$= n(a\theta + b)^{n-1} a$$

$\rightarrow b$ is constant

$$\frac{\partial J(\theta)}{\partial \theta_0} = \frac{1}{N} \sum_{i=0}^{N-1} (\theta_0 + \theta_1 x_i - y_i) = \frac{1}{N} (\theta_0 + \theta_1 x_i - y_i)$$

$$\frac{\partial J(\theta)}{\partial \theta_1} = \frac{1}{N} \sum_{i=0}^{N-1} (\theta_0 + \theta_1 x_i - y_i) (x_i) \\ = \frac{1}{N} \sum_{i=0}^{N-1} (\theta_0 + \theta_1 x_i - y_i) x_i$$

Impact of learning rate

α - Alpha

- Too small learning rate (α) - gradient descent is

slow to optimize.

- Too large learning rate (α) gradient descent may fail to converge due to large updates to parameter value for every gradient.

Types of Linear Regression

① Simple Linear Regression $Y = \theta_0 + \theta_1 x_1 + \epsilon$

It consists of single independent variable (feature) and single dependent variable (output)

E.g.: House Price prediction based on carpet area of house.

② Multiple Linear Regression $Y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \epsilon$

It consists of multiple independent variables (features) and single dependent (output) variable

E.g.: House Price prediction based on carpet area of house, city, locality, distance to best school etc.

③ Multivariate Linear Regression

It consists of multiple independent variables (features) and multiple dependent variables (output)

E.g.: Temperature and humidity based on city, season, month, rainfall.

How to find parameters.

Gradient Descent Optimization (Most used)

Ordinary Least Squares (Applicable to simple linear regression as well)

Ordinary Least Squares

$$Y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots$$

It is a common/ordinary method used for finding the best fitting line through a set of data points by minimizing the sum of the squared differences between the observed and predicted values.

Note: As this method involves matrix operations such as inverse computation its computationally and memory expensive method for high dimensional data.

$$Y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 \approx \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 \Big|_{x_0=1}$$

$$Y = X\theta \text{ (Matrix Notation)}$$

$$\theta = (X^T X)^{-1} X^T Y.$$

Why to introduce x_0 ? $\rightarrow$ Algebraic Equation

$$y = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2$$

x_0	x_1	x_2	y
x_{01}	x_{11}	x_{21}	y_1
x_{02}	x_{12}	x_{22}	y_2
x_{03}	x_{13}	x_{23}	y_3

$$\hat{y}_1 = \theta_0 x_{01} + \theta_1 x_{11} + \theta_2 x_{21}$$

$$\hat{y}_2 = \theta_0 x_{02} + \theta_1 x_{12} + \theta_2 x_{22}$$

$$\hat{y}_3 = \theta_0 x_{03} + \theta_1 x_{13} + \theta_2 x_{23}$$

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \hat{y}_3 \end{bmatrix} = \begin{bmatrix} \theta_0 x_{01} + \theta_1 x_{11} + \theta_2 x_{21} \\ \theta_0 x_{02} + \theta_1 x_{12} + \theta_2 x_{22} \\ \theta_0 x_{03} + \theta_1 x_{13} + \theta_2 x_{23} \end{bmatrix}$$

Converting to matrix notation

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \hat{y}_3 \end{bmatrix}_{3 \times 1} = \begin{bmatrix} x_{01} & x_{11} & x_{21} \\ x_{02} & x_{12} & x_{22} \\ x_{03} & x_{13} & x_{23} \end{bmatrix}_{3 \times 3} \times \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix}_{3 \times 1}$$

$$\Rightarrow \boxed{\hat{y} = X\theta} \rightarrow \text{Matrix Equation.} \quad \hat{y} \rightarrow y \text{ Predicted.}$$

$$\text{Since } \hat{y} = X\theta$$

$$\Rightarrow y = X\theta \Rightarrow \theta = y$$

$$\Rightarrow \theta = x^T y$$

$$\Rightarrow \theta = x^T y \cdot \frac{x^T}{x^T}$$

$$\Rightarrow \boxed{\theta = (x^T x)^{-1} x^T y} \rightarrow \text{Involves very costly Mathematical Computations.}$$

(Resource Intensive)

Evaluation of Regression Models.

① MAE

Mean Absolute Error

It is average of absolute error values

$$MAE = \frac{1}{M} \sum_{i=1}^M |y_i - \hat{y}_i|$$

Why absolute error?

∴ Positive errors will be balanced by negative errors

$$\begin{array}{l} y_1 \quad \hat{y}_1 \quad y_1 - \hat{y}_1 = e_1 \\ y_2 \quad \hat{y}_2 \quad y_2 - \hat{y}_2 = e_2 \\ y_3 \quad \hat{y}_3 \quad y_3 - \hat{y}_3 = e_3 \\ y_4 \quad \hat{y}_4 \quad y_4 - \hat{y}_4 = e_4 \end{array}$$

② MAPE

Mean Absolute Percentage Error

MAE Represented in Percentage format

$$MAPE = \frac{1}{M} \sum_{i=1}^M |(y_i - \hat{y}_i) / y_i| * 100$$

③ MSE

Mean Square Error

$$MSE = \frac{1}{M} \sum_{i=1}^M (y_i - \hat{y}_i)^2$$

Why Square?

To balance the -ve & +ve errors.

④ RMSE

Root Mean Square Error.

It is Square root of average squared Error values.

$$RMSE = \sqrt{\frac{1}{M} \sum_{i=1}^M (y_i - \hat{y}_i)^2} = \sqrt{MSE} = \text{SQRT}(MSE)$$

⑤ $R^2 \rightarrow R$ Square

- Evaluation Metric for Regression model
- Tells you how much performance gain is provided by new model over the base model.

For purpose of R^2 , base model always predicts the average value for every row.

x_1	x_2	y	Base Model Prediction	New Model Prediction	Error Base	Error New Model
x_{01}	x_{02}	y_0	$\bar{y} = \frac{1}{n} \sum_{i=0}^n y_i$	$\hat{y}_0$	$y_0 - \bar{y}$	$y_0 - \hat{y}_0$
x_{11}	x_{12}	y_1	$\bar{y}$	$\hat{y}_1$	$y_1 - \bar{y}$	$y_1 - \hat{y}_1$
x_{21}	x_{22}	y_2	$\bar{y}$	$\hat{y}_2$	$y_2 - \bar{y}$	$y_2 - \hat{y}_2$
x_{31}	x_{32}	y_3	$\bar{y}$	$\hat{y}_3$	$y_3 - \bar{y}$	$y_3 - \hat{y}_3$

$$MSE_{\text{Base}} = \frac{1}{M} \sum_{i=0}^{n-1} (y_i - \bar{y})^2$$

$$MSE_{\text{New}} = \frac{1}{M} \sum_{i=0}^{n-1} (y_i - \hat{y}_i)^2$$

$$\text{Then } R^2 = 1 - \frac{MSE_{\text{New}}}{MSE_{\text{Base}}}$$

⑤ $R^2 \rightarrow$ R Square / Coefficient of determination

- Evaluation Metric for Regression model

- Tells you how much performance gain is provided by new model over the base model.

For purpose of R^2 , base model always predicts the average value for every row.

x_1	x_2	y	Base Model Prediction	New Model Prediction	Error Base	Error New Model
x_{01}	x_{02}	y_0	$\bar{y} = \frac{1}{M} \sum_{i=0}^{M-1} y_i$	$\hat{y}_0$	$y_0 - \bar{y}$	$y_0 - \hat{y}_0$
x_{11}	x_{12}	y_1	$\bar{y}$	$\hat{y}_1$	$y_1 - \bar{y}$	$y_1 - \hat{y}_1$
x_{21}	x_{22}	y_2	$\bar{y}$	$\hat{y}_2$	$y_2 - \bar{y}$	$y_2 - \hat{y}_2$
x_{31}	x_{32}	y_3	$\bar{y}$	$\hat{y}_3$	$y_3 - \bar{y}$	$y_3 - \hat{y}_3$

$$MSE_{Base} = \frac{1}{M} \sum_{i=0}^{M-1} (y_i - \bar{y})^2$$

$$MSE_{New} = \frac{1}{M} \sum_{i=0}^{M-1} (y_i - \hat{y}_i)^2$$

Then

$$R^2 = 1 - \frac{MSE_{New}}{MSE_{Base}}$$

R Squared Score : It is the proportion of the variation in the dependent variable that is predicted from the independent variables

It is also called coefficient of determination

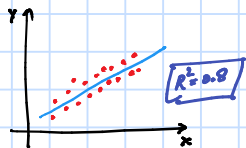
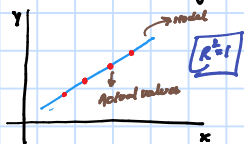
$\Rightarrow$ Mean of y $\bar{y} = \frac{1}{M} \sum_{i=0}^{M-1} y_i$

$\Rightarrow$ Sum of squares of Residuals : $SS_{res} = \frac{1}{M} \sum_{i=0}^{M-1} (y_i - \hat{y}_i)^2$

$\Rightarrow$ Total Sum of Squares : $SS_{tot} = \frac{1}{M} \sum_{i=0}^{M-1} (y_i - \bar{y})^2$

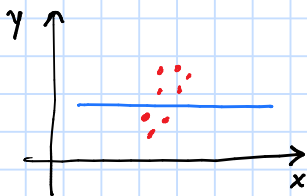
$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

R^2 will be 1 for model which predicts values which exactly match actual values.



R^2 will be 0 for model which predicts values as mean of y (baseline).

$$R^2 = 0 = 1 - \frac{MSE_{New}}{MSE_{Base}} \quad \text{Such that} \quad MSE_{New} = MSE_{Base}$$



R^2 will be negative for model which has worst prediction than baseline.

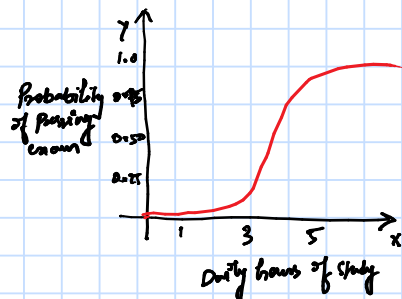
$$R^2 < 0$$

$$\Rightarrow 1 - \frac{MSE_{New}}{MSE_{Base}} < 0$$

$$\Rightarrow MSE_{New} > MSE_{Base}$$

Logistic Regression

- It is a supervised learning algorithm - labelled data points are needed to train a model.
- Logistic Regression is a classification model.
- It predicts the probability of event.
- Output is in range of 0 and 1.



Examples

- # Predict whether student will pass in exam based on hours of study.
- # Predict whether email is spam or not.

Why Regression if it's a classification algorithm?
⇒ Because the output is in probability from 0 to 1.

In the given chart

- x axis represents hours of study, and y axis represents probability of passing the exam.
- If probability $P \geq 0.5 \rightarrow$ Student will pass the exam.
- If probability $< 0.5 \rightarrow$ Student will not pass the exam.

→ This is an example of "binary classification" → two classes

In order to classify we need threshold which by default is 50% / 0.5

0.5 is the threshold.

↓
Mostly defined by business goals.

→ Logistic Regression gives the probability of an event taking place by having the log-odds for the event be a linear combination of one or more independent variables.

• Linear Regression $\rightarrow h_\theta(x) = \theta_0 + \theta_1 x$

• Logistic Regression $\rightarrow g(h_\theta(x))$

Where $g(z) = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-(\theta_0 + \theta_1 x)}}$

Log Odds is a general concept for probability & statistics.

odds?

odds = $\frac{\text{Probability of Success}}{\text{Probability of Failure}}$

If probability of success = P

then Probability of Failure = $1-P$

$\Rightarrow \text{Odds} = \frac{\text{Probability of Success}}{\text{Probability of Failure}} = \frac{P}{1-P}$

Hence $\log \text{Odds} = \log \frac{P}{1-P}$

Logistic Regression models log odds i.e. $\log \frac{P}{1-P}$ as a linear combination of features.

$\Rightarrow \boxed{\log \frac{P}{1-P} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots} \rightarrow$

considering single variable in equation

$\Rightarrow \log \frac{P}{1-P} = \theta_0 + \theta_1 x \Rightarrow \frac{P}{1-P} = e^{\theta_0 + \theta_1 x}$

$\Rightarrow P = (1-P)e^{\theta_0 + \theta_1 x} \Rightarrow P = e^{\theta_0 + \theta_1 x} - P e^{\theta_0 + \theta_1 x}$

$\Rightarrow e^{\theta_0 + \theta_1 x} = P + P e^{\theta_0 + \theta_1 x}$

$\Rightarrow P(1 + e^{\theta_0 + \theta_1 x}) = e^{\theta_0 + \theta_1 x} \Rightarrow P = \frac{e^{\theta_0 + \theta_1 x}}{1 + e^{\theta_0 + \theta_1 x}}$

$\Rightarrow P = \frac{e^{\theta_0 + \theta_1 x}}{e^{\theta_0 + \theta_1 x} \left(\frac{1}{e^{\theta_0 + \theta_1 x}} + 1 \right)} \Rightarrow P = \frac{1}{1 + \frac{1}{e^{\theta_0 + \theta_1 x}}}$

$\Rightarrow P = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$

If $z = \theta_0 + \theta_1 x$

$\Rightarrow \boxed{P = \frac{1}{1 + e^{-z}}}$

↓
Basic formula for the logistic expression

$$P = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}} \quad \text{Logistic Function}$$

$$P = f(\theta_0 + \theta_1 x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

$$f(0) = \frac{1}{1 + e^0} = \frac{1}{1 + 1} = \frac{1}{2} = 0.5$$

$$f(\infty) = \frac{1}{1 + e^{-\infty}} = \frac{1}{1 + 0} = \frac{1}{1} = 1$$

$$\boxed{e^0 = 1}$$

$$f(-\infty) = \frac{1}{1 + e^{\infty}} = \frac{1}{1 + \infty} = \frac{1}{\infty} = 0$$

$$\boxed{e^{\infty} = \infty}$$

∞ represents large positive number

$\Rightarrow$ You input any x into $f(x) \Rightarrow$ Output will be a number between 0 and 1 that can be interpreted as probability.

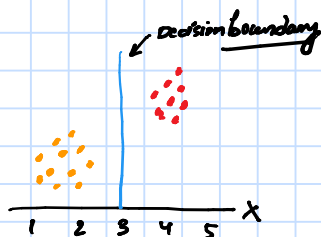
$\Rightarrow$ Output values from logistic equation will be in the range of 0 to 1.

- If $x = 0$ then $f(x) = 0.5$
- If $x > 0$ then $f(x) > 0.5$
- If $x < 0$ then $f(x) < 0.5$

Output be between 0 and 1.

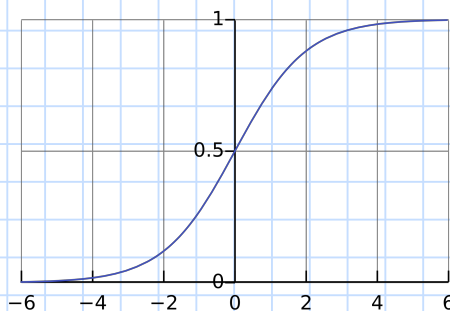
DECISION BOUNDARY

- Decision boundary represents the dividing line or surface that separates different classes in a feature space.
- It defines the region where the model predicts one class or label on one side and another class or label on the other side.



- Decision boundary will always be in terms of feature values.

- Threshold will always be in terms of probability.



- LOGISTIC REGRESSION ONLY & ONLY LINEAR DECISION BOUNDARIES
- IT CANNOT MAKE QUADRATIC, CUBIC ETC.
- IT WON'T ACCEPT ANY NON-LINEAR RELATIONSHIPS.

Decision boundary is the feature value at probability reaching threshold

Decision boundary for 2 features - Linear

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

• Logistic Regression $g(h_{\theta}(x))$

$$\Rightarrow P = g(h_{\theta}(x)) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x_1 + \theta_2 x_2)}}$$

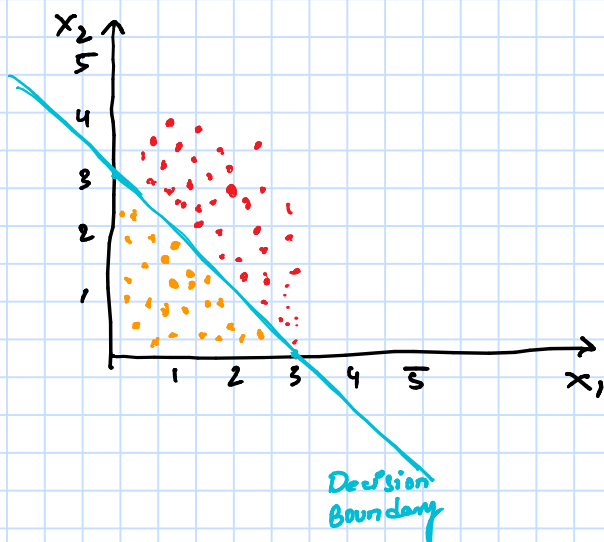
Consider, $\theta_0 = -3$, $\theta_1 = 1$ and $\theta_2 = 1$

$$\begin{aligned}\Rightarrow h_{\theta}(x) &= \theta_0 + \theta_1 x_1 + \theta_2 x_2 \\ &= -3 + (1)x_1 + (1)x_2 \\ &= -3 + x_1 + x_2\end{aligned}$$

Since the logistic function has output between 0 & 1

$$\Rightarrow x_1 + x_2 - 3 \geq 0 \Rightarrow x_1 + x_2 \geq 3$$

If $x_1 + x_2 - 3 \geq 0$ then $g(h_{\theta}(x)) \geq 0.5$.
Predicted class is class 1



Pb

If

$$\theta_0 = -4$$

$$\theta_1 = 2$$

$$\theta_2 = 2$$

$$\begin{aligned}\Rightarrow \theta_0 + \theta_1 x_1 + \theta_2 x_2 \\ &= -4 + 2x_1 + 2x_2 \\ &= -2 + x_1 + x_2\end{aligned}$$

$$x_1 + x_2 - 2 \geq 0$$

$$\boxed{x_1 + x_2 \geq 2}$$

② is the decision boundary

Decision Boundary - Non Linear

How to introduce linearity in Logistic Regression?

→ By using feature engineering.

Example:

x_1	x_2	$x_3 = (x_1)^2$	$x_4 = (x_2)^2$
x_{10}	x_{20}	$(x_{10})^2$	$(x_{20})^2$
x_{11}	x_{21}	$(x_{11})^2$	$(x_{21})^2$
x_{12}	x_{22}	$(x_{12})^2$	$(x_{22})^2$
x_{13}	x_{23}	$(x_{13})^2$	$(x_{23})^2$

Now the decision boundary will be

$$\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4 x_4$$

$$= \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 (x_1)^2 + \theta_4 (x_2)^2$$

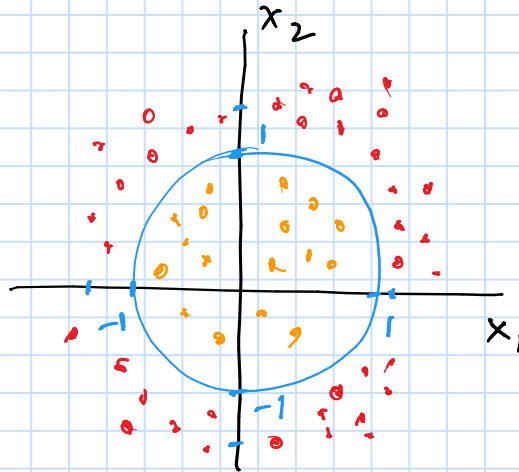
Consider

$$\theta_0 = -1, \theta_1 = 0, \theta_2 = 0, \theta_3 = 1, \theta_4 = 1$$

$$\Rightarrow \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 (x_1)^2 + \theta_4 (x_2)^2$$

$$= -1 + x_1^2 + x_2^2 \Rightarrow x_1^2 + x_2^2 = 1$$

$$\Rightarrow \boxed{x_3 + x_4 = 1}$$



Basic Equation of a circle

$$(x-h)^2 + (y-k)^2 = r^2$$

$(h, k) \rightarrow$ Co-ordinates of the center of the circle.

$r \rightarrow$ Radius of the circle.

$(x, y) \rightarrow$ any point lying on the circle.

If Center of circle $(0, 0)$

then Equation of circle is

$$\underline{x^2 + y^2 = r^2}$$

The decision boundary of logistic regression is inherently a linear one.

Through feature engineering or polynomial transformations, logistic regression can capture non-linear relationships between features and the target variable.

$$\left| \begin{array}{l} x_1 \rightarrow x_1^2 \\ x_1 \rightarrow x_1^3 \end{array} \right|$$

COST FUNCTION.

For Linear Regression,

$$\text{Cost Function} = \frac{1}{M} \sum_{i=1}^M e_i^2$$

$$= \frac{1}{M} \sum_{i=1}^M (y_i - \hat{y}_i)^2$$

x_1	x_2	y	Predicted Prob
x_{10}	x_{20}	1	0.99
x_{11}	x_{21}	0	0.001
x_{12}	x_{22}	0	0.002
x_{13}	x_{23}	1	0.95
x_{14}	x_{24}	1	0.85

Historical Data

x_1 - Daily hours of study

x_2 - Attendance.

y - Actual label (Binary)

Errors - Break this error into 2 parts

Part 1: $y_{\text{actual}} = 1$

$$\text{error} = -\log(\text{Predicted Probability})$$

negative sign is included because log of number less than 1 is negative

$$\log(\text{number} < 1) = -ve$$

$$\log(0.1) = -1 \text{ thus } -\log(0.1) = -(-1) = 1.$$

⇒ If Predicted probability is close to 1 then $-\log(\text{prob})$ will approach 0

i.e. if P 's close to 1 (0.99)

$$\text{then } -\log(P) \rightarrow 0$$

⇒ If Predicted probability is close to 0, then $-\log(P) \rightarrow \infty$] Not ideal

$$\hookrightarrow \log(P \rightarrow 0) \rightarrow -\infty$$

Part 2. When $y_{\text{actual}} = 0$

$$\text{Error} = -\log(1-P)$$

$P \rightarrow$ Predicted Probability

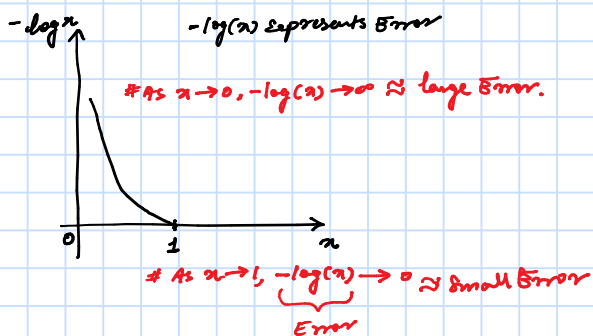
$$\text{If } P_{\text{prob}} \rightarrow 0, \text{ Error} = -\log(1-0)$$

$$= -\log(1) = 0$$

$$\text{If } P_{\text{prob}} \rightarrow 1, \text{ Error} = -\log(1-0.99999) = -\log(0.00001)$$

$$= -(-5) = 5$$

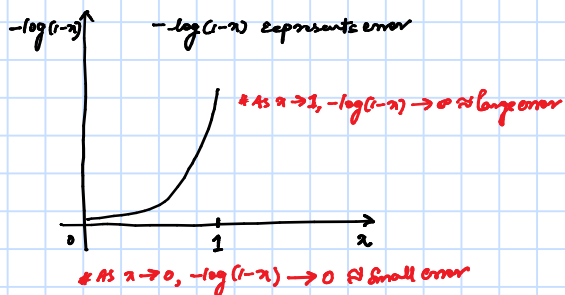
For Error = $-\log(P)$



$$\log(1) = 0$$

$$\log(0) = -\infty$$

For Error = $-\log(1-P)$



Logarithmic loss is well-suited for measuring the difference between predicted probabilities and actual class labels.

It encourages the model to assign high probabilities to the correct class and low probabilities to the incorrect class.

Combining the Errors

$$\begin{aligned}\text{Error} &= -\log(h_0(x)) \quad Y=1 \\ &\quad -\log(1-h_0(x)) \quad Y=0\end{aligned}$$

Cost function (Log-likelihood loss)

$$J(\theta) = -Y * \log(g(h_0(x))) - (1-Y) * \log(1-g(h_0(x)))$$

where

$$g(h_0(x)) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

Parameter Estimation :

Gradient descent optimization.

Evaluation of Classification Models //

① $\text{Accuracy} = (\text{Correct classifications}) * 100 / (\text{All classifications})$
Value lies in the range of 0 to 100%

In case of imbalanced dataset check accuracy of prediction for each class.

Otherwise even if classifier predicts all the samples belong to majority class - accuracy will be high.

Imbalanced Datasets :

Imbalanced datasets have unequal proportions of positive and negative
 $\underbrace{\hspace{1cm}}_{Y=1}$ and $\underbrace{\hspace{1cm}}_{Y=0}$

② Confusion Matrix

> Assuming Binary classification

Actual y (Y)	Predicted y ($\hat{y}$)	
	1	0
1	2 (TP)	3 (FN)
0	3 (FP)	2 (TN)

1. TP: True Positive
2. FN: False Negative
3. FP: False Positive
4. TN: True Negative

Actual	Predicted	
	Positive	Negative
Positive	20 12 (TP)	8 (FN)
Negative	80 10 (FP)	70 (TN)

Positive - has cancer
Negative - Do not have cancer

Actual	Prob Prob	$\hat{y}$
1	0.9	1
1	0.8	1
0	0.7	1
0	0.6	1
0	0.52	1
1	0.4	0
1	0.3	0
1	0.2	0
0	0.1	0
0	0.01	0

Example

Considering following confusion matrix for case of cancer predicting model,

① TPR: True Positive Rate = $\frac{\text{True Positive}}{\text{Total Positive}} = \frac{TP}{P} = \frac{12}{20} = \frac{3}{5} = 0.6$

② TNR: True Negative Rate = $\frac{\text{True Negative}}{\text{Total Negative}} = \frac{TN}{N} = \frac{70}{80} = \frac{7}{8} = 0.875 = 0.88$

③ FPR: False Positive Rate = $\frac{\text{False Positive}}{\text{Total Negative}} = \frac{FP}{N} = \frac{10}{80} = \frac{1}{8} = 0.125 = 0.13$

④ FNR: False Negative Rate = $\frac{\text{False Negative}}{\text{Total Positive}} = \frac{FN}{P} = \frac{8}{20} = \frac{2}{5} = 0.4$

	TPR	TNR	FPR	FNR
IDEAL MODEL	1	1	0	0
IDEAL IMPERFECT MODEL	0	0	1	1

③ Precision, Recall & F1 score

Binary classification confusion matrix.

① Precision = $\frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} = \frac{TP}{(TP+FP)}$
 $= \frac{2}{(2+3)} = \frac{2}{5} = 0.4 = 40\%$

* Precision means out of positive predictions how many were actually positive.

② Recall = $\frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} = \frac{TP}{(TP+FN)}$
 $= \frac{2}{(2+3)} = \frac{2}{5} = 0.4 = 40\%$

* Recall tells you out of actual positives, how many were predicted correctly.

* Recall is same as TPR.

③ F1 Score: F1 score is the harmonic mean of Precision and Recall.

$\Rightarrow F1 \text{ score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

Actual	Predicted	
	1	0
1	2 (TP)	3 (FN)
0	3 (FP)	2 (TN)

Precision & Recall values depend upon threshold for classification

* High threshold will give high precision but low recall.

* Classification threshold should be chosen as per business domain - how model will be used, What's the impact of missing true positive prediction etc.

Q Evaluate the model using following predictions at thresholds 0.5, 0.4 & 0.75

Y Actual	Predicted Probability	$\hat{y}$ (0.5)	$\hat{y}$ (0.4)	$\hat{y}$ (0.75)
1	0.9	1	1	1
1	0.8	1	1	1
0	0.7	1	1	0
0	0.6	1	1	0
0	0.52	1	1	0
1	0.4	0	1	0
1	0.3	0	0	0
0	0.2	0	0	0
1	0.1	0	0	0
0	0.01	0	0	0

Confusion matrix at threshold 0.4/40%

	TP	FN	Predicted
Actual	1	0	
	3	2	
	FP	TN	
	3	2	

$$TPR = TP/P = 3/3+2 = 3/5 = 0.6 = 60\%$$

$$TNR = TN/N = 2/3+2 = 2/5 = 0.4 = 40\%$$

$$FPR = FP/N = 3/3+2 = 3/5 = 0.6 = 60\%$$

$$FNR = FN/P = 2/3+2 = 2/5 = 0.4 = 40\%$$

$$Precision = TP/TP+FP = 3/3+3 = 3/6 = 1/2 = 0.5 = 50\%$$

$$Recall = TP/P = TP/TP+FN = 3/3+2 = 3/5 = 0.6 = 60\%$$

$$F1-Score = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

$$= \frac{2 \times 0.5 \times 0.6}{0.5 + 0.6}$$

$$= \frac{2 \times 8 \times 6}{10 \times 6} \times \frac{16}{8 \times 6} = \frac{2}{10} = 0.2 = 20\%$$

Confusion matrix at threshold 0.5/50%

	TP	FN	Predicted
Actual	1	0	
	2	3	
	FP	TN	
	3	2	

$$TPR = \frac{TP}{P} = \frac{2}{5} = 0.4$$

$$TNR = \frac{TN}{N} = \frac{2}{5} = 0.4$$

$$FPR = \frac{FP}{N} = \frac{3}{5} = 0.6$$

$$FNR = \frac{FN}{P} = \frac{3}{5} = 0.6$$

$$Precision = TP/TP+FP = \frac{2}{2+3} = \frac{2}{5} = 0.4 = 40\%$$

$$Recall = TP/P = \frac{2}{2+3} = \frac{2}{5} = 0.4 = 40\%$$

$$F1-Score = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

$$= \frac{2 \times 0.4 \times 0.4}{0.4 + 0.4} = \frac{2 \times 4 \times 4}{10 \times 6} \times \frac{16}{8 \times 6} = \frac{4}{10} = 0.4 = 40\%$$

Confusion matrix at threshold 0.75

	TP	FN	Predicted
Actual	1	0	
	2	3	
	FP	TN	
	0	5	

$$TPR = TP/P = \frac{2}{5} = 0.4 = 40\%$$

$$TNR = TN/N = \frac{5}{5} = 1 = 100\%$$

$$FPR = FP/N = 0$$

$$FNR = FN/P = \frac{3}{5} = 0.6 = 60\%$$

$$Precision = TP/TP+FP = \frac{2}{2+0} = 1 = 100\%$$

$$Recall = TP/P = \frac{2}{2+3} = \frac{2}{5} = 0.4 = 40\%$$

$$F1-Score = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

$$= \frac{2 \times 1 \times 0.4}{1 + 0.4} = \frac{2 \times 4}{10} \times \frac{16}{147} = \frac{4}{7} = 0.57 = 57\%$$

Q How to translate business goals into the evaluation metrics in logistic regression?

Can we say $\rightarrow$ We are either trying to reduce False Positives or False Negatives at any given time?

Are precision & recall inversely proportional to each other?

As we increase threshold, the false positive decreases
hence precision increases and Recall decreases.

As we decrease threshold, the precision decreases
and Recall increases.



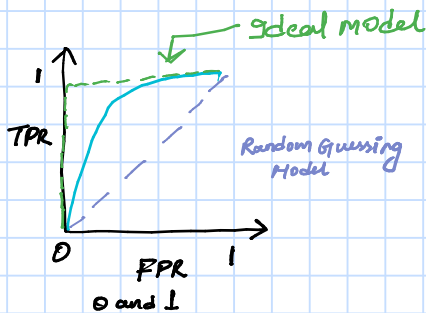
④ ROC curve

Receive Operating Characteristics (ROC)

Is the graph showing the Performance of a classification model at all classification thresholds.

This curve plots two Parameters

- True Positive Rate (TPR)
- False Positive Rate (FPR)



Step 1: Take multiple threshold values
[0.1, 0.2, 0.3, ..., 0.9,] etc.

For each threshold value generate a confusion matrix
and calculate TPR & FPR

Step 2: Plot TPR against FPR

ROC - Graph is qualitative

AUC $\rightarrow$ Area under the curve $\rightarrow$ Quantitative.

* AUC for ideal curve = 1

* AUC for random guessing model = 0.5

* AUC closer to 1, Better the model.

* AUC less than 0.5, worse model.

$$0.5 < AUC \leq 1$$

Multiclass classification

Multiclass classification is needed when we must predict whether data point belongs to one of the n -classes where $n > 2$

- ⇒ Predict the month based on weather parameters - 12 classes.
- ⇒ Predict whether email belongs to family, work or friends - 3 classes.
- ⇒ Predict whether image belongs to bicycle, cat or scooter - 3 classes.

11 One-vs-All (Trick)

No. of classes = 3

class 0

class 1

class 3

The trick is to convert this into a binary classification problem.

Dataset

F_1	F_2	Y
-	-	0
-	-	1
-	-	2
-	-	0
-	-	1
-	-	2
-	-	1
-	-	2

→ This dataset will be transformed into 3-datasets.

How?

- Convert multiclass problem into multiple binary problems.

class 0
class 1
class 2 } → convert into 3 binary classification problems.

- ① - class 0 vs No class 0
- ② - class 1 vs No class 1.
- ③ - class 2 vs No class 2

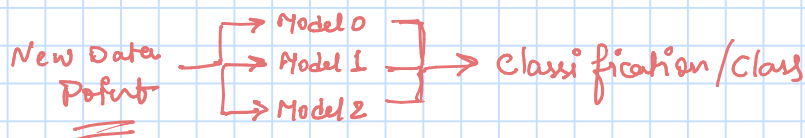
• Train n -logistic regression models/classifiers where n is number of classes

- Each classifier/model classifies exactly 1 class vs all other classes learning to separate each class from rest of the classes.

① Model 0

	F_1	F_2	Y
class 0	-	-	1
	-	-	0
	-	-	0
	-	-	1
	-	-	0
	-	-	0
	-	-	0
	-	-	0

- For prediction, select the class that gives the highest probability.



Model Regularization

- Underfitting and Overfitting
- Bias-Variance trade-off
- L1-Regularization
- L2-Regularization.

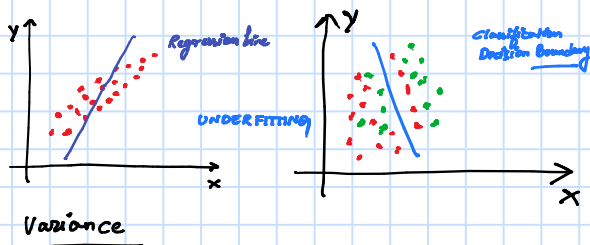
Underfitting $\rightarrow$ Not able to capture the trend.

Overfitting $\rightarrow$ capturing each and every point.

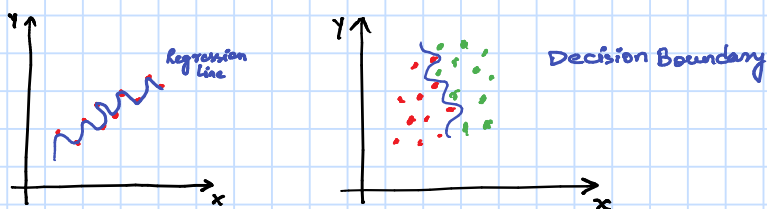
Optimal fitting $\rightarrow$ Able to capture the general trend.

Bias

- Bias represents inability of machine learning algorithm to capture the true relationship between the data points.
- Bias is the error introduced by approximating a real world problem with a simplified model.
- Models with high bias are called underfitted models.

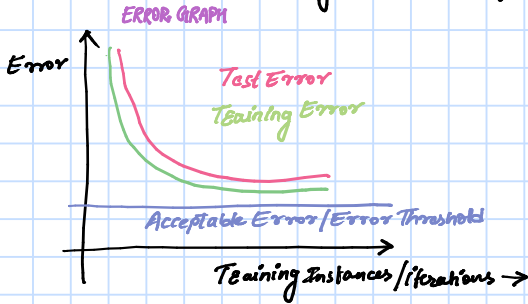


- Variance is inability of machine learning model to generalize learnings from training data.
- Variance is the error introduced by the model's sensitivity to variations in the training data.
- Models with high variance can predict accurate results on training set but have high error on test set or unseen data.
- This is the result of over fitting on the training data.



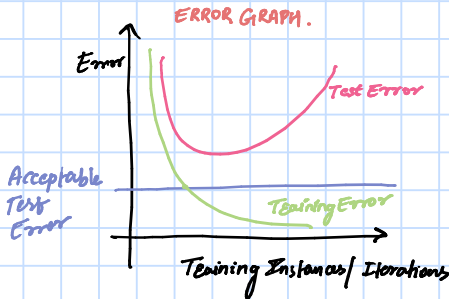
Bias Symptoms

- Models with high Bias have high training error and high error on test as well.
- If the current features are not able to capture all the factors contributing to response variable then it will result in high Bias (underfitting).



Variance Symptoms

- Models with high Variance give low error on training data but high error on testing data.
- Highly complex model with large number of features can result in high Variance models.



How to reduce Bias:

- Capture more data.
- Feature Engineering
- Increase the complexity
 - Introduce more features
- Change the algorithm.

How to reduce model variance:

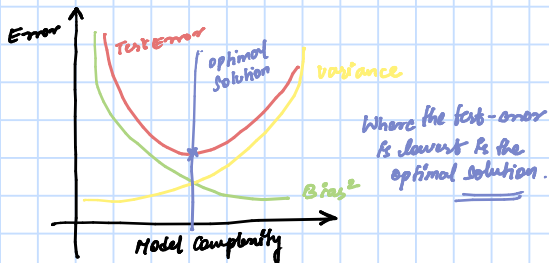
- Reduce input features.
- Reduce number of parameters in model (use less complex model).
- Increase training data.
- Introduce Regularization.

How to reduce model Bias:

- Explore other potential features by discussing with business stakeholders.
- Better optimization of model parameters (another algorithm).
- Increase number of iterations and/or training data.
- Create new features by using current features
 - In house price prediction, convert room sizes into carpet area.
 - In case of classifying engine as defective or non-defective using production date and sell date to create number of months on shelf as a new feature.

Bias Variance Trade off.

- As model increases variance tends to increase and bias decreases.
- Models with less complexity generally result in high bias.



* Regularization

- Regularization is the method of creating simpler (less complex) models.
- It is achieved by adding penalty term to the cost function.
- It helps in reducing complexity of model by reducing some feature coefficients (θ_1, θ_2) etc. towards zero.
- Multiple Linear Regression Cost function with regularization

$$J(\theta) = \frac{1}{2M} \sum_{i=1}^M (\theta_0 + \theta_1 x_{i1} + \theta_2 x_{i2} - y_i)^2 + \lambda R(\theta)$$

- λ - Controls contribution of regularization function to Cost function.
- $R(\theta)$ is the regularization function chosen to impose penalty.
- Total Penalty = $\lambda R(\theta)$.

Cost Function = Total Error + $\lambda \times$ Penalty Term.

Penalty Term $\propto$ Complexity of the model $R(\theta)$

$\lambda \rightarrow$ Decides how much to focus on penalty term.

If λ is small then focus is on Total Error

If λ is large then focus is on penalty term $R(\theta)$

$\lambda \rightarrow$ Hyper Parameter.

* L1 Regularization

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$

$$R(\theta) = \sum_{i=1}^M |\theta_i| = |\theta_1| + |\theta_2| + |\theta_3|$$

$$\text{Cost Function } J(\theta) = \text{Total Error} + \lambda (|\theta_1| + |\theta_2| + |\theta_3|)$$

By decreasing the cost function $\rightarrow$ Decreasing the penalty
Decreasing the $|\theta|$ parameters

Generally the λ is kept at 0.2 or 0.01

NOTE: θ_0 is not included

decreasing $|\theta|$ parameters implies we are reducing or decreasing the magnitude of θ parameters.

* Complexity of a model can be measured by the number of θ parameters and the magnitude of parameters.

* L1 Regularization is typically used when there are large number of features.

* L1 regularization adds a penalty equal to the absolute value of the magnitude of coefficients.

* This type of model can result in sparse models with fewer features. $\rightarrow$ Features are automatically removed.

* Some coefficients can become zero and eliminated from the model (less important features)

* Multiple Linear Regression Cost function in L1 Regularization also called Lasso Regression:

$$J(\theta) = \frac{1}{2M} \sum_{i=1}^M (\theta_0 + \theta_1 x_{i1} + \theta_2 x_{i2} - y_i)^2 + \lambda \sum_{i=1}^M |\theta_i|$$

$|\theta_i|$ represents absolute value of coefficients.

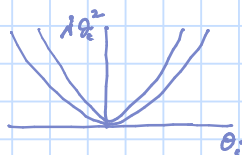
L2 Regularization

- # L2 regularization adds penalty equal to the square of the magnitude of coefficients.
- # This type of regularization does not result in sparse models with fewer coefficients.
- # Features are not eliminated from the model.
- # Also called RIDGE REGRESSION.
- # Multiple linear regression cost function with L2 Regularization:

$$J(\theta) = \frac{1}{2M} \sum_{i=1}^M (\theta_0 + \theta_1 x_{i1} + \theta_2 x_{i2} - y_i)^2 + \frac{\lambda}{2} \sum_{j=1}^n |\theta_j|^2$$

- $|\theta_j|^2$ represents square of the magnitude of coefficients.
- In this case n will be 2 as we have used two coefficients.

NOTE: This type of regularization is typically used when there are large number of features.



$$\begin{aligned}\theta_j &\leftarrow \theta_j - \eta \frac{\partial (\text{Cost Function})}{\partial \theta_j} \\ \theta_j &\leftarrow \theta_j - \eta \left[\frac{\partial (\text{Error})}{\partial \theta_j} + \frac{\partial (10_j^2)}{\partial \theta_j} \right] \\ \theta_j &\leftarrow \theta_j - \eta \left[\frac{\partial (\text{Error})}{\partial \theta_j} + 2\lambda \theta_j \right] \\ \text{Assume } \theta_j \text{ is not useful feature} \\ \frac{\partial (\text{Error})}{\partial \theta_j} &= 0 \\ \theta_j &\leftarrow \theta_j - \underbrace{2\eta \lambda \theta_j}_{\text{Dependent}}\end{aligned}$$

$\lambda \rightarrow$ Hyperparameter provided by user.

Key points in Regularization

Regularization can be applied to regression as well as classification models

- Regularization is useful when there are large number of features.
- Regularization helps in avoiding overfitting.
- L2 regularization can be used for feature selection.
- Regularization is useful in presence of multicollinear features.

Significance of λ :

- Very high value of λ results in higher penalty - less complex model thus lean towards underfitting.
- Very low value of λ results in very low penalty thus leaning towards overfitting - No benefit of regularization

Clustering

- unsupervised learning.
- Only features are present and no labels.
- We focus only on finding the patterns in data.
- It automatically identifies group of similar examples.

E.g.: Clustering Models

Principal Component Analysis (PCA)

Q How do we measure the similarity between the data points?

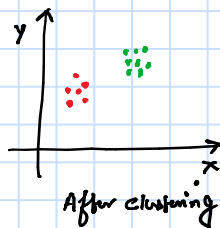
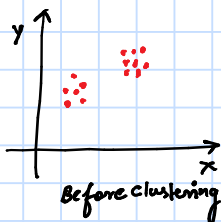
- In a cluster all the datapoints are similar to each other.
- All the data points in one cluster are dissimilar to all the datapoints in other cluster.

What is clustering?

- clustering is grouping of data points in such a way that similar data points are in same group and dissimilar data points are in different groups.

Use Cases of clustering:

- Customer Segmentation - Identify customers with similar interests.
- Identify themes in customer complaints.



K-Means Clustering

- K means clustering is method of clustering data points into K clusters (K is input to the algorithm)

• Distance metric: Euclidean Distance

• Algorithm Steps:

- 1. Initialization $\rightarrow$ Random Initialization of k-centroids
- 2. Assignment $\rightarrow$ Assign datapoints to nearest centroid based on Euclidean distance.
- 3. Update $\rightarrow$ Update every cluster centroid by averaging all the data points in that cluster.

Continue Assignment and update until:

- cluster centroids do not change further
- specified number of iterations completed.

$$R_i = (10, 20)$$

$$R_j = (20, 10)$$

$$R_k = (5, 6)$$

$$\begin{aligned} d(R_i, R_j) &= \sqrt{(10-20)^2 + (20-10)^2} \\ &= \sqrt{(-10)^2 + (10)^2} \\ &= \sqrt{100+100} = \sqrt{200} = 13.43 \end{aligned}$$

$$\begin{aligned} d(R_i, R_k) &= \sqrt{(10-5)^2 + (20-6)^2} \\ &= \sqrt{(5)^2 + (14)^2} \\ &= \sqrt{25+196} = \sqrt{221} \\ &= 14.87 \end{aligned}$$

Since $d(R_i, R_j) < d(R_i, R_k)$

thus R_i & R_j are similar to each other

while R_i & R_k are dissimilar to each other.

Steps:

① Initialization:

- we have to provide the number of clusters to make
e.g. $k=3$
- we have to initialize the centroids $\rightarrow$ 3 centroids
 $\downarrow$
Random Initialization.

Goal is to provide some random data points (as in above example). These data points can be some of the rows or they can be not from the data rows.

$k \rightarrow$ Hyperparameter $\rightarrow$ Input to the algorithm.

② Assignment:

- Assign each data point to the nearest centroid.
- we use Euclidean Distance

In above example, since we initialized with 3 centroids thus we will have to calculate Euclidean distance of each data point/row from the 3 randomly initialized points, and assign data points to the centroid having smallest distance.

③ Update $\rightarrow$ Update of centroids

- to calculate the new centroid, use average within the centroid.
- Now Iterate for n-number of times and repeat step ② & ③ till we reach the Convergence:

We either reach pre-decided number of iterations,
or

There is no change in the assignment of data points.
or

There is no or very little change in the centroids.

1 Optimization Objective

* k-means optimization objective minimizes within-cluster variance (squared Euclidean distances).

• Optimization Objective:
Cost Function

Also called
RSS $\rightarrow$ Residual Sum of Squares.

Minimize $\frac{1}{M} \sum_{i=1}^M \|x_i - \mu_{ci}\|^2$ $\rightarrow$ we have to minimize cost function given the value of k or given the number of clusters.

$\rightarrow M$: Number of data points to be clustered.

$\rightarrow x_i$: d dimensional data point

$\rightarrow \mu_{ci}$: Centroid of cluster to which datapoint x_i is assigned

c is in range 1 to k where k is the number of clusters.

NOTE: Optimization might converge at local minima.

Need of Feature Scaling.

- Example

• 'Hours of study' feature is in range of 0 to 5

• 'Classroom attendance' feature is in range 0 to 100.

- As k-means clustering uses Euclidean distance metric it will suffer if we have features with different scales.

• In this example if we apply clustering without scaling then classroom attendance feature will dominate all the clustering with very low importance to the hours of study.

Solution: Scale features to equal scale whenever possible.

- In this example the features can be scaled to 0 to 1.

How to decide the value of k ? $\rightarrow$ The value of k is decided based on some experiments.

$\rightarrow$ Decide a range of values $k = [2, 3, \dots, 20]$

$\rightarrow$ Run k-means for various numbers of k i.e. for every value in chosen range run k-means clustering.

$\rightarrow$ Plot cost function against the number of clusters i.e. every k .

Key points to remember

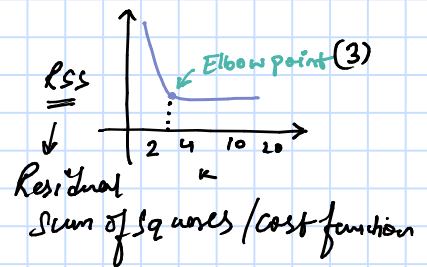
* Elbow points does not guarantee best results.

* In many cases a clear elbow point won't be found.

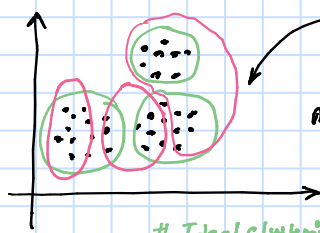
* Choose the number of clusters based on business objective and evaluate clustering results based on the same.

* Understand how clustering results would be used.

* Check if stakeholders have any idea about number of groups/clusters they are expecting.



Effect of Initialization of Centroids on k-means



How to deal with this?

Method 1: Run k-means clustering for multiple times for same value of k for example $k=3$ 10 times:

1	$k=3$	RSS_1
2	$k=3$	RSS_2
3	$k=3$	RSS_3
...		
10	$k=3$	RSS_{10}

then choose the model minimum RSS .

Ideal clustering
could happen if we are not smart about initialization.

Method 2: Also called K-Means++

Enhanced versions of K-means.

- Only step 1 is different, step 2 and step 3 are exactly same.

let's say we have to initialize 3 centroids; $k=3$

Step 1: 1. Choose 1st Centroid randomly

2. Choose 2nd Centroid which is farthest away from Centroid 1.

3. Choose 3rd Centroid which is farthest away from Centroid 1 and 2.

	F_1	F_2	D_1	D_2	D_{min}
R_1			d_{11}	d_{12}	$\min(d_{11}, d_{12})$
R_2			d_{21}	d_{22}	$\min(d_{21}, d_{22})$
R_3			d_{31}	d_{32}	$\min(d_{31}, d_{32})$
R_4			d_{41}	d_{42}	$\min(d_{41}, d_{42})$
R_5			d_{51}	$d_{52}=0$	$\min(d_{51}, d_{52})$
R_6			d_{61}	d_{62}	$\min(d_{61}, d_{62})$
					$\max(D_{min})$

D_1 Distance from centroid 1

D_2 Distance from centroid 2

• d_{51} - distance from R_5 chosen centroid R_5

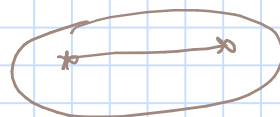
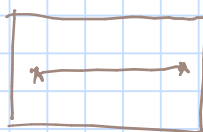
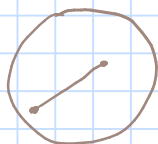
• $d_{52}=0$ because it is the same point as R_5 and its distance from itself will be 0.

Chose the row which has the highest D_{min} i.e. $\max(D_{min})$

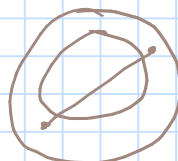
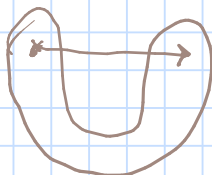
Demerits of K-means clustering

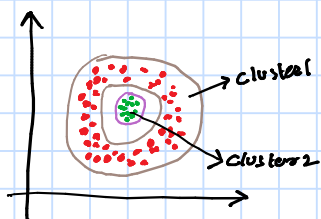
- ① You have to provide the value of k
- ② k-means clustering does not deal with the outliers automatically.
- ③ It can only make convex shape clusters.

Convex Shapes



Concave Shapes





DBSCAN

⇒ Density Based Spatial Clustering
OF Applications with Noise.

* It can automatically handle outliers/noise.

* Algorithm works on the concept of density in the feature space.

Q How the density is defined?

Q How the outliers are defined?

Q How the outliers are handled?

* It groups points that are closely packed together (points with many nearby neighbors).

and points that lie alone in low density regions are marked as outliers.

* Distance Metric: Euclidean, Cosine, Hamming, Mahalanobis etc.

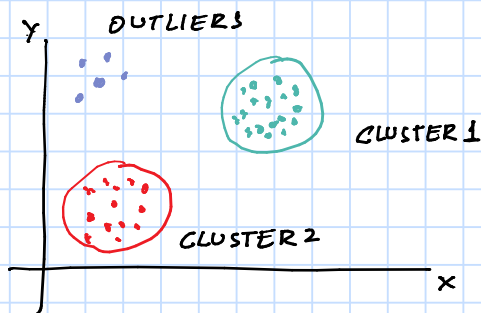
* Parameters:

- ϵ (Epsilon): Distance threshold to consider a point as neighbor.

- minPts: Minimum number of points needed in neighborhood to form a cluster.

* If number of neighbors more than minPts then dense region.

* If number of neighbors less than minPts then not dense region.



Algorithm Steps

① Find the points around (neighbors) every point with in distance ϵ (Eps)

② we give labels to each point:

- Core point
- Non Core point
- Outlier

Label points as core points if they have more than minPts neighbors (including the point itself)

A point is a core point if it has $\geq \text{minPts}$ as neighbors.

A non-core point is a point having $< \text{minPts}$ neighbors but has a core point as a neighbor.

An outlier is a point which has $< \text{minPts}$ neighbors and do not have core point as neighbor.

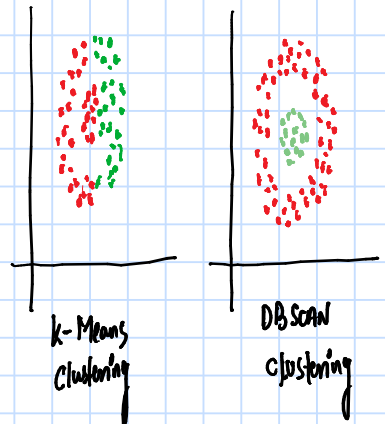
③ Connect core points (consider them as belong to the same cluster) if they are within ϵ (Epsilon) distance from each other.

④ Assign each non-core point to a nearby cluster if any point in cluster is within distance ϵ (Epsilon) neighbor, otherwise assign it to noise (Outliers).

NOTE: Because a non-core point has more than one core point as neighbor i.e. within the Epsilon distance, then it will be connected to the core point which is CLOSEST.

Advantages of DBSCAN Clustering

- If data domain is well understood then ' ϵ ' and ' minPts ' can be chosen appropriately for clustering, no need to specify number of clusters.
- Works well even for clusters with arbitrary shape.
- It can identify and separate outliers instead of just adding them to nearest cluster.
- We can use distance metrics other than Euclidean distance.



HIERARCHICAL CLUSTERING → Offers flexible clustering.

• Hierarchical clustering method builds a hierarchy of clusters.

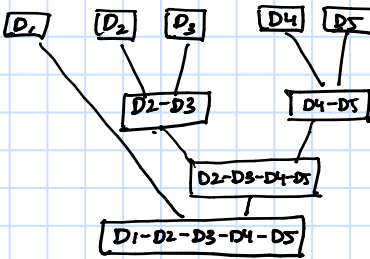
• Types of Hierarchical clustering:

① Agglomerative: Each observation starts its own cluster, and pairs of clusters are merged based on distance metric and 'linkage criteria'.

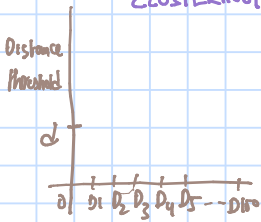
② Divisive: All observations start as one cluster and splits are performed recursively based on distance metric and 'linkage criteria'.

ISSUES

• Distance metric: Euclidean, Cosine, Manhattan etc. ① How to calculate distance between 2 data points? → Distance Metric



AGGLOMERATIVE CLUSTERING



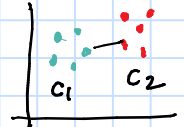
d -distance threshold to classify data points in same cluster

• If number of data points = 100 and $d = 0$ then total number of clusters will be 100.

• if $d \rightarrow \infty$ then the number of clusters will be 1.

① How to calculate distance between a data point and a cluster? → Linkage Criteria

② How to calculate distance between 2 clusters?



Linkage Criteria decides the condition on which points are merged (agglomerative) into same cluster.

I. Single Linkage: It is the distance between the closest points of two clusters.

Q. What is the distance between C1 and C2?

Ans. The distance between the closest points of the clusters.

II. Complete Linkage: It is the distance between the farthest points of two clusters.

III. Average Linkage: Distance between each pair of data points in two clusters is calculated.

Total distance is divided by number of data points in two clusters to get average value.

III Ward linkage:

- This linkage criteria tries to minimize sum of squared distance within cluster.
- Two clusters are merged together or points are assigned to cluster based on how much it will increase within the cluster sum of squared distance from centroid.
- This method is least susceptible to noise and outliers.

* Single linkage criteria is most susceptible to noise.



SSD - Sum of Squared Distances.

- hypothetically imagine that we merged C1 and C2 into C12



then we calculate SSD

$$\Delta SSD_{12} = SSD_{12} - (SSD_1 + SSD_2)$$

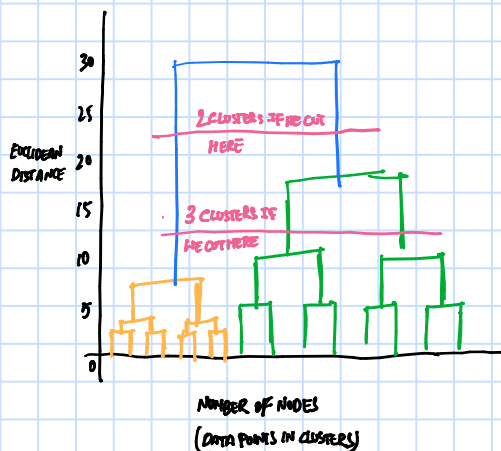
$$\Delta SSD_{13} = SSD_{13} - (SSD_1 + SSD_3)$$

$$\Delta SSD_{23} = SSD_{23} - (SSD_2 + SSD_3)$$

- * Merge two clusters for which ΔSSD is minimum.
- * For outliers or noise ΔSSD will be very-very high thus the clusters won't be merged.

* Dendrograms.

- * Dendrograms are tree-like structures which show the way in which data is split into groups based on distance.
- * It is useful in deciding number of clusters.
- * Dendrogram is cut at a point based on distance threshold to get clusters.



NOTE Dendrograms are good only for small number of points.

Number of clusters

Option 1:

- Number of clusters can be decided by visualizing dendrogram and cutting the right point.
 - But it is very difficult with large real world datasets.
- Dendrograms are only good for small set of data points.

Option 2:

- Decide a right linkage criteria and distance metric based on dataset
- Provide distance threshold parameter (agglomerative clustering implemented in scikit-learn)
- Above this threshold data points/clusters won't be changed.

Option 3:

- Create number of clusters as per business requirement (Keep this in mind even if any other method is used)

Advantages:

- Number of clusters need not be specified
- Options to choose from various distance metrics based on type of data (e.g cosine distance can be used for text data)
- Distance threshold suitable for business domain can be chosen to get optimized results for the business requirement.

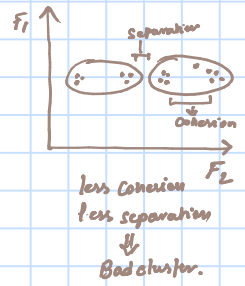
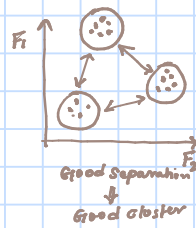
Disadvantages:

- Slow speed of execution for large datasets.
- Large memory consumption makes it unsuitable for very large datasets.

Evaluation Methods For Clustering.

A good clustering is the one where:

- ① Similar data points are grouped together $\rightarrow$ Cohesion.
- ② Dissimilar data points are in different groups $\rightarrow$ Separation.



Silhouette Coefficient.

looks at both separation and Cohesion.

Silhouette Coefficient is calculated for every data point.

Silhouette Coefficient for every data point is defined as:

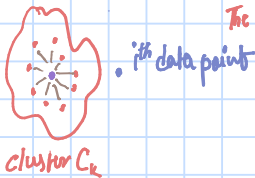
$$S(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

$i \rightarrow$ i th data point

$a(i)$: Average intracluster distance
 $b(i)$: Average distance to all points in the nearest cluster.

$\Rightarrow a(i)$: Average intra cluster distance represents Cohesion

Assume the i th data point in dataset belongs to cluster C_k



The average intra cluster distance

$$a(i) = \frac{1}{|C_k|} \sum_{j \in C_k, j \neq i} d_{ij}$$

$|C_k| \rightarrow$ Number of data points in cluster C_k

higher the average intracluster distance lower the Cohesion

$\Rightarrow b(i)$: Average distance to all the points in the nearest cluster.

$b(i)$ represents Separation.



$$b(i) = \frac{1}{|C_n|} \sum_{j \in C_n} d_{ij}$$

higher the $b(i)$ larger the Separation.

$\Rightarrow$ If you have 100 data points

$S(1), S(2), S(3), \dots, S(100)$

calculate $S(i)$ for every point

then take average Silhouette Coefficient

$$\langle S \rangle = \frac{1}{M} \sum_{i=1}^M S(i) \quad M=100$$

$$\langle S \rangle = \underline{-1 \text{ to } +1} \quad +1: \text{ Ideal value}$$

$$S(i) = \frac{b(i) - a(i)}{\max(b(i), a(i))}$$

Case I: $b(i) > a(i)$

$$\Rightarrow S(i) = \frac{b(i) - a(i)}{b(i)} = 1 - \frac{a(i)}{b(i)}$$

Case II: $b(i) < a(i)$

$$\Rightarrow S(i) = \frac{b(i) - a(i)}{a(i)} = \frac{b(i)}{a(i)} - 1$$

Ideal cluster would be one for which $b(i) \gg a(i)$

$\Rightarrow$ Ideal value of $S(i)$ for a good cluster will be near 1.

Worst cluster would be one for which $a(i) \gg b(i)$

$\Rightarrow$ The value of $S(i)$ will be -1 .

Thus $S(i)$ will range from -1 to 1

$-1 \rightarrow$ Worst

$+1 \rightarrow$ Ideal value.

How to use Silhouette Score for k-means clustering

In k-means clustering we used to plot RSS (Residual Sum of Squares) against the number of clusters and then locate the elbow point to decide the ideal number of clusters.



⇒ But in the cases where finding elbow point is difficult we can use $SC(i)$ to decide the ideal number of clusters.

Let's assume k ranges from 3 to 10, then we calculate Silhouette Score at each value of k .

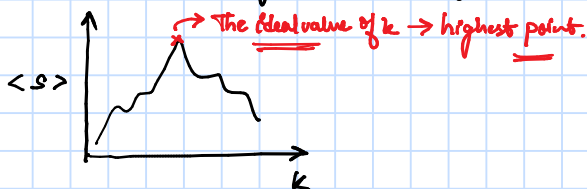
at $k=3 \rightarrow \langle S_3 \rangle$

$k=4 \rightarrow \langle S_4 \rangle$

$\vdots$

$k=10 \rightarrow \langle S_{10} \rangle$

then we plot Silhouette Scores against the values of k .



4 Practical Evaluation

Even though Silhouette is a good statistical metric, in practical scenarios it might not be useful.

You can use one/both of the following methods to evaluate clustering

Results.

- Manual inspection of random samples from large clusters
→ Are they capturing similar pattern?
- Create small labelled data of samples which should be in same cluster and samples which should never be in same cluster and check in which clusters these labelled data points lie after clustering.

Pre-clustering